Do Now · retrieval

Lesson 154 · Search Engines: Crawling, Indexing and Retrieval

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Why does a server recheck a browser form?

Q2

When can binary search be used?

Q3

What data structure follows LIFO?

Q4

What is a database foreign key?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Discover — Crawler fetches pages found through links and sitemaps.

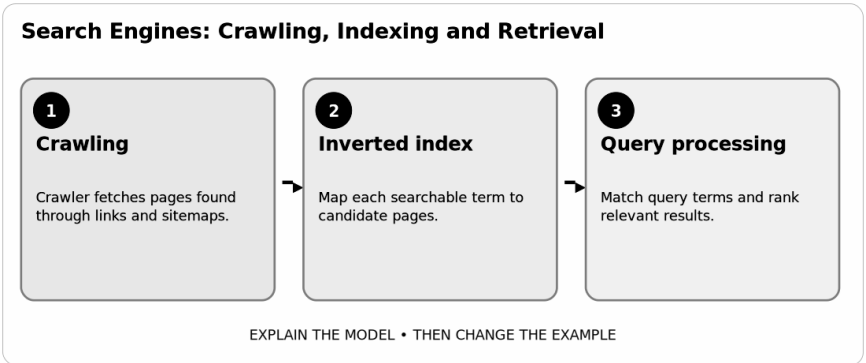
Core idea 2

Index — Map each searchable term to candidate pages.

Core idea 3

Retrieve — Match query terms and rank relevant results.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

crawler — Search agent fetching linked pages.

inverted index — Terms mapped to document identifiers.

tokenisation — Breaking text into searchable units.

ranking — Ordering candidate search results.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Page A has 'binary search' and Page B 'binary tree'. Build a small inverted index.

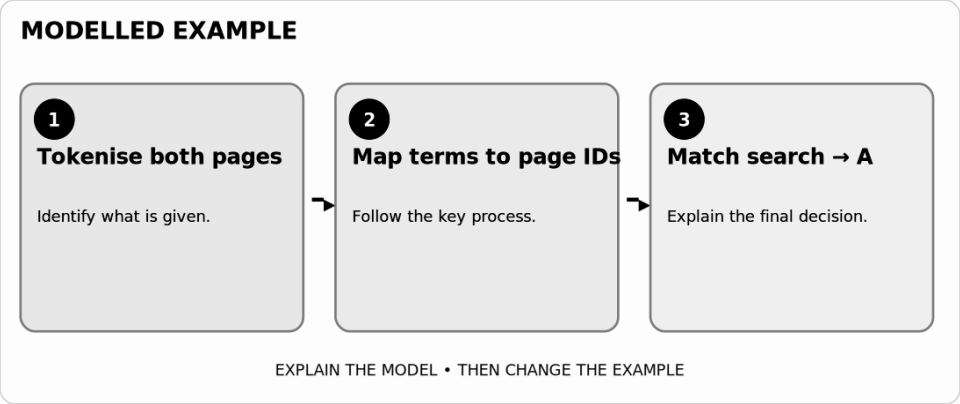
Follow the worked method

1. Tokenise both pages — Identify the input or diagram and predict the first result.

2. Map terms to page IDs — Follow the second stage and explain its importance.

3. Match search → A — binary→A,B; search→A; tree→B. The query 'binary search' intersects both postings and retrieves A.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Index terms

Identify the input or diagram and predict the first result.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Explain crawling and indexing. [6]

Question 2

Index P='cat dog', Q='dog bird' and identify result for 'dog bird'. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Discuss why a new page might not appear in search and why robots.txt is not a security control. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Search indexing and result ranking are the same operation.

Use key vocabulary from memory

1. Explain crawler in your own words.

2. Explain inverted index in your own words.

3. Explain tokenisation in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.